



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-25

Experiment 25. Prolog Program for Fruit and Its Color Using Backtracking

Aim

To write a Prolog program to find the **color of fruits using backtracking**.

Objectives

- To understand backtracking in Prolog.
- To store fruit-color relationships.
- To retrieve multiple matching results.

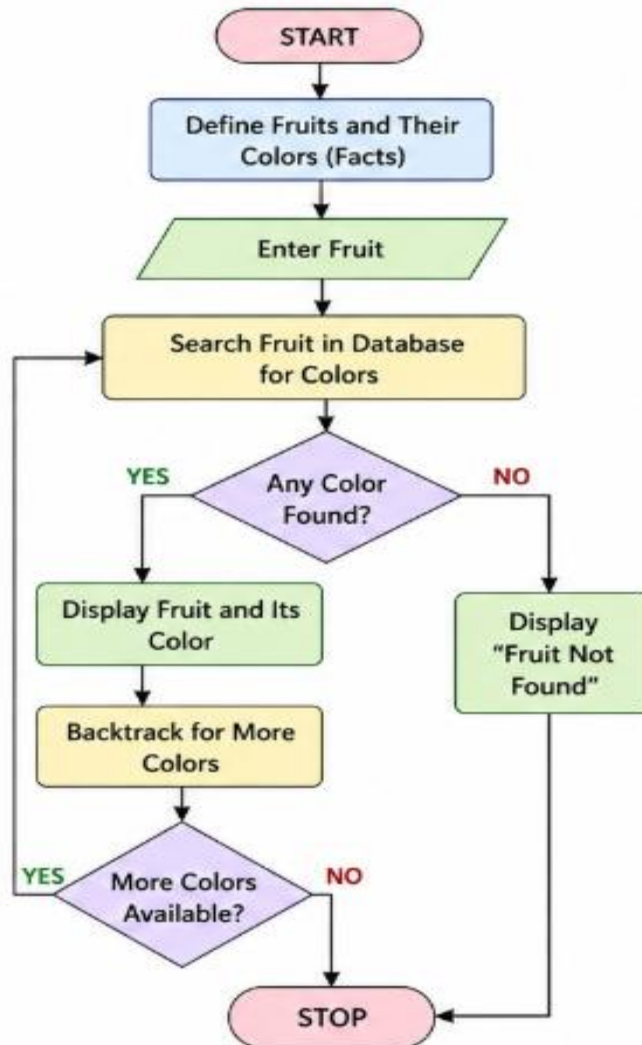
Algorithm

1. Define fruits and their colors.
2. Select a fruit.
3. Search for its color.
4. Prolog backtracks to find additional matching facts.
5. Display the results.

Flowchart

25. PROLOG PROGRAM FOR FRUIT AND ITS COLOR USING BACKTRACKING

FLOWCHART



Code

fruit(apple, red).

fruit(apple, green).

fruit(banana, yellow).

```
fruit(grape, green).
```

```
fruit(orange, orange).
```

```
main :-
```

```
    Fruit = apple,
```

```
    fruit(Fruit, Color),
```

```
    write('Fruit: '),
```

```
    write(Fruit),
```

```
    write(' Color: '),
```

```
    write(Color), nl,
```

```
    fail.
```

```
main.
```

```
:- initialization(main).
```

Output

OneCompiler 

 AI  Prolog  Run 

HelloWorld.pl ×

```
1 fruit(apple, red).
2 fruit(apple, green).
3 fruit(banana, yellow).
4 fruit(grape, green).
5 fruit(orange, orange).
6
7 main :-
8     Fruit = apple,
9     fruit(Fruit, Color),
10    write('Fruit: '), write(Fruit),
11    write(' Color: '), write(Color), nl,
12    fail.
13
14 main.
15
16 :- initialization(main).
```

I/O  AI Agent

STDIN
Input for the program (Optional)

Output:
Fruit: apple Color: red
Fruit: apple Color: green

Result

Fruit: apple Color: red

Fruit: apple Color: green

Conclusion

Thus, the fruit and color relationship was successfully implemented using Prolog with backtracking.